The Battle Plan: From Assembly to Shellcode

1. **Master the Linux Environment & Stack Mechanics:** Estimated time: 1-2 weeks of high-intensity CLI work. Skip dry OS textbooks. Learn the memory layout directly through terminal challenges:

- **Target 1: OverTheWire (Bandit & Narnia)**: Finish Bandit to get comfortable with Linux permissions, SSH, redirects, and pipes. Move immediately to Narnia for basic C buffer overflows.
- **The Core Concepts to Lock Down**: The x86-64 stack frame, registers (`RSP`, `RBP`, `RIP`, `RAX`, `RDI`, `RSI`), endianness, and how the `call`, `leave`, and `ret` instructions manipulate memory.

2. **Grind pwn.college (The Golden Standard):** The single best hands-on computer systems curriculum on earth. Head to **pwn.college** (built by Yan Shoshitaishvili and the CTF team at Arizona State University). It is completely free, interactive, and runs in live Docker containers.

- Work through the modules in strict order: **Program Misuse** → **Assembly Crash Course** → **Debugging** → **Shellcode Injection** → **Memory Errors**.
- This directly replaces everything you missed in a formal CS degree, but teaches it from the attacker's perspective.

3. **Build Your Arsenal (GDB + GEF + Pwntools + Ghidra):** Weaponize your Python skills. Set up your native Linux environment (or a dedicated Ubuntu VM / WSL2 setup) with the standard offensive toolkit:

- **GEF (GDB Enhanced Features)**: Turns standard GDB into a dynamic visualization tool showing registers, stack memory, and instructions at every breakpoint.
- **Pwntools (`pip install pwntools`)**: The de facto Python framework for exploit development. Write scripts that automate process communication, calculate cyclical buffer offsets (`cyclic()`), pack addresses (`p64()`), and craft interactive payloads.
- **Ghidra**: Your primary static analysis suite for decompiling native binaries, inspecting symbol tables, and tracing xrefs.

4. **Build Verifiable Proof-of-Work:** How you turn skill into income and credibility without a resume. To earn bounties or land elite security roles without credentials, your public paper trail must do the talking:

- **Writeups**: When you solve hard challenges on pwn.college, HackTheBox, or CTFs, write clean, technical post-mortems on GitHub Pages or a Markdown blog explaining the exact root cause, stack trace, and exploit payload.
- **Author Crackmes**: Build your own x64 challenges with custom anti-debugging tricks, register obfuscation, or virtual machine packers and publish them back to Crackmes.one.
- **Open Source Vulnerability Research**: Target small open-source C/C++ projects or IoT firmware on GitHub using fuzzers (like AFL++ or Honggfuzz). Finding one real memory corruption bug and filing a CVE (Common Vulnerabilities and Exposures) is worth ten college degrees.
